

MPA Cosmology Seminar, 09.04.24

Testing dark photon dark matter with intergalactic gas



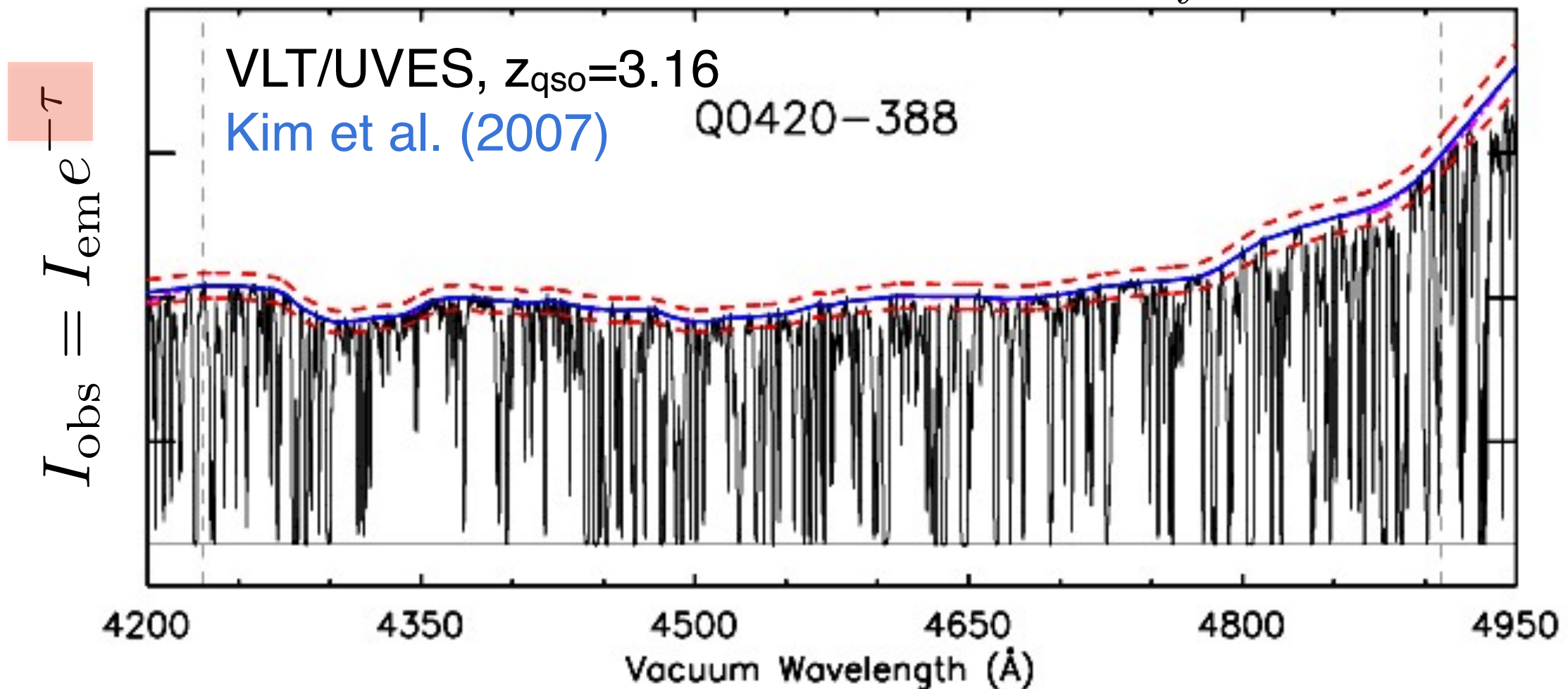
Jamie Bolton

With thanks to:

Hongwan Liu (NYU), Andrea Caputo (CERN), Matteo Viel (SISSA),
Andrea Trost (SISSA)

The Lyman- α forest

$$\lambda_{\text{Ly}\alpha} = 1216 \text{ \AA}$$



Optical depth

$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T_0^{0.7} H(z) \Gamma(z)} (1+\delta)^{2-0.7(\gamma-1)}$$

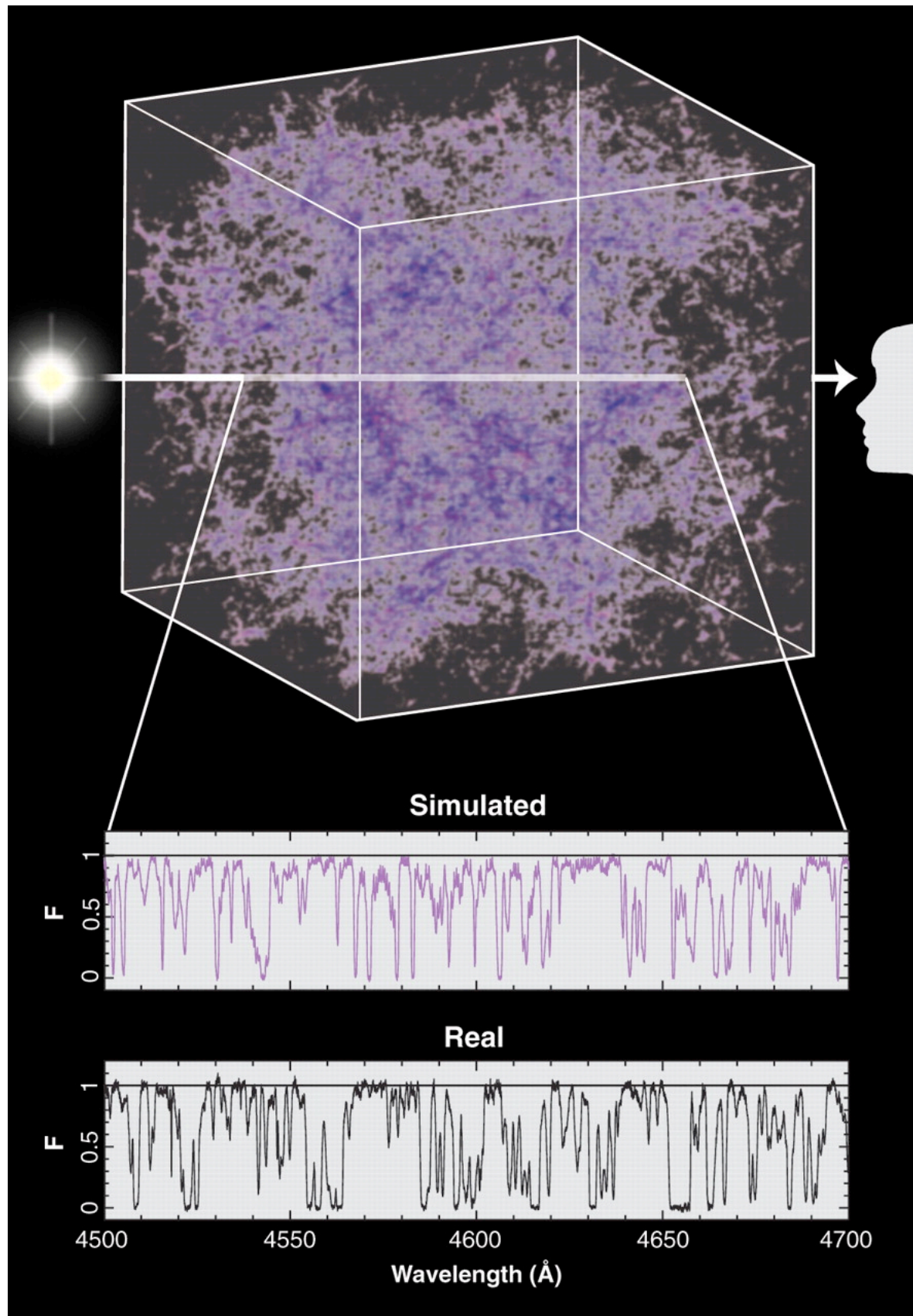
Temperature

Density

Hydrogen photo-ionisation rate

Basic IGM recipe (first done in 1990s)

- Cosmological parameters
 - Matter power spectrum, $P(k)$
 - (Newtonian) gravity
 - Hydrodynamics (>100 kpc/h)
 - Uniform photo-ionisation and heating (hydrogen and helium only)
 - Integrate w.r.t time (lots...)
- ⇒ Lyman- α forest (+ IGM in general)



Faucher-Giguere et al. (2008)



Sherwood-Relics

Hybrid radiation-hydrodynamical models of the Lyman-alpha forest.

Looking for the original Sherwood? Click [here](#).



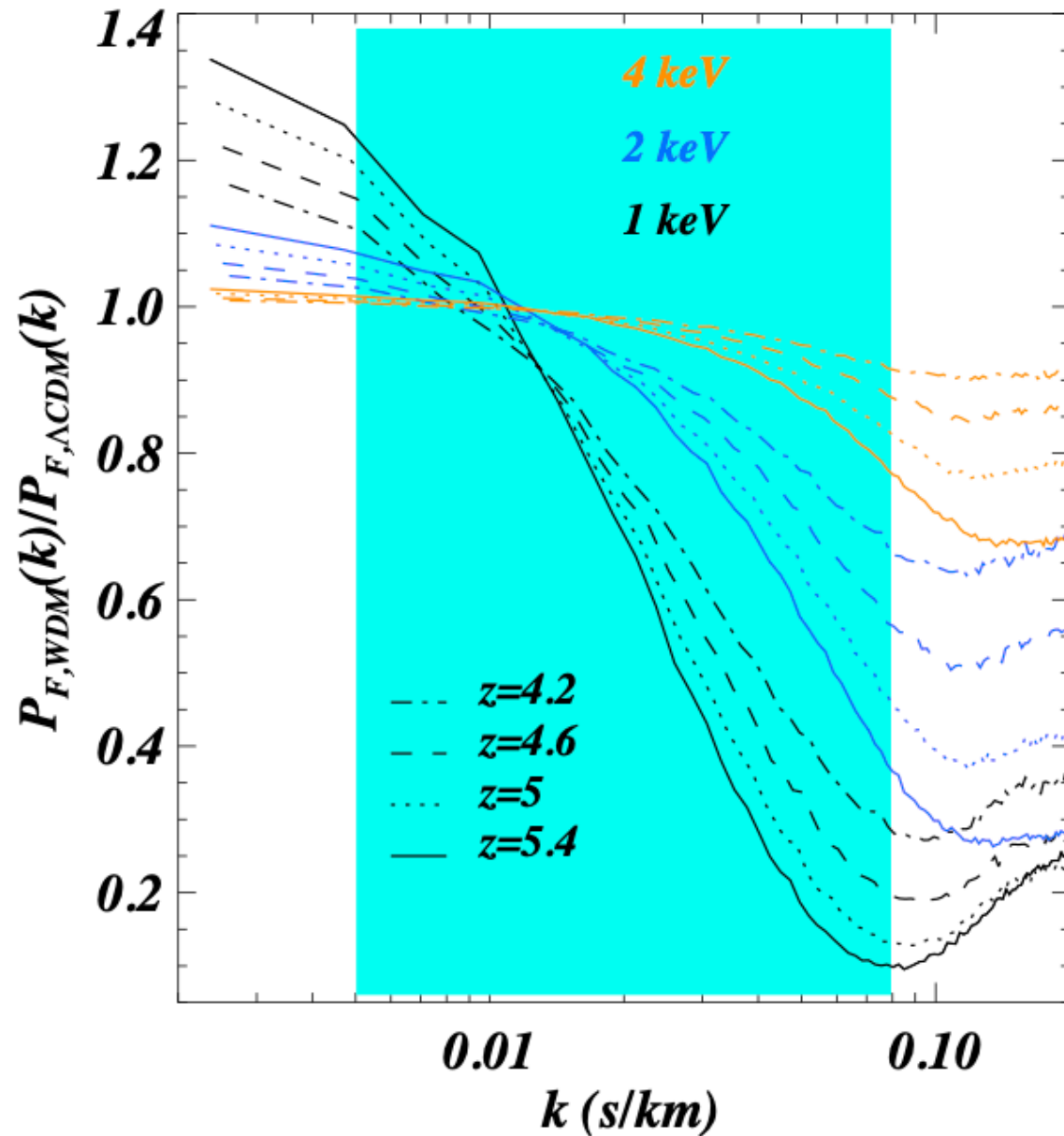
Science & Technology
Facilities Council



DiRAC

<http://www.nottingham.ac.uk/astronomy/sherwood-relics/>

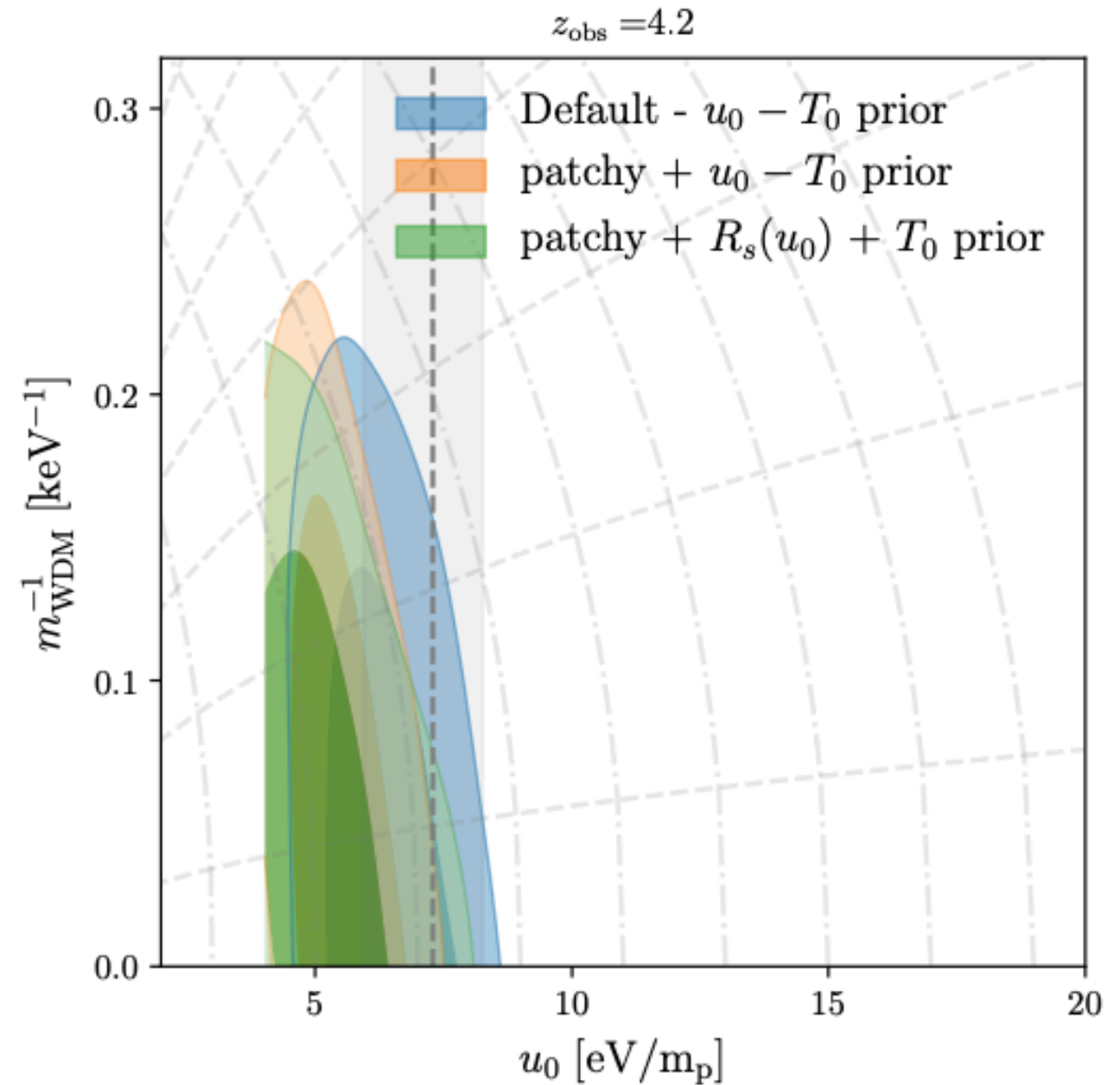
Small scale implications for WDM



Viel et al. (2013)

No small scale suppression from reionisation

Including scale suppression from reionisation

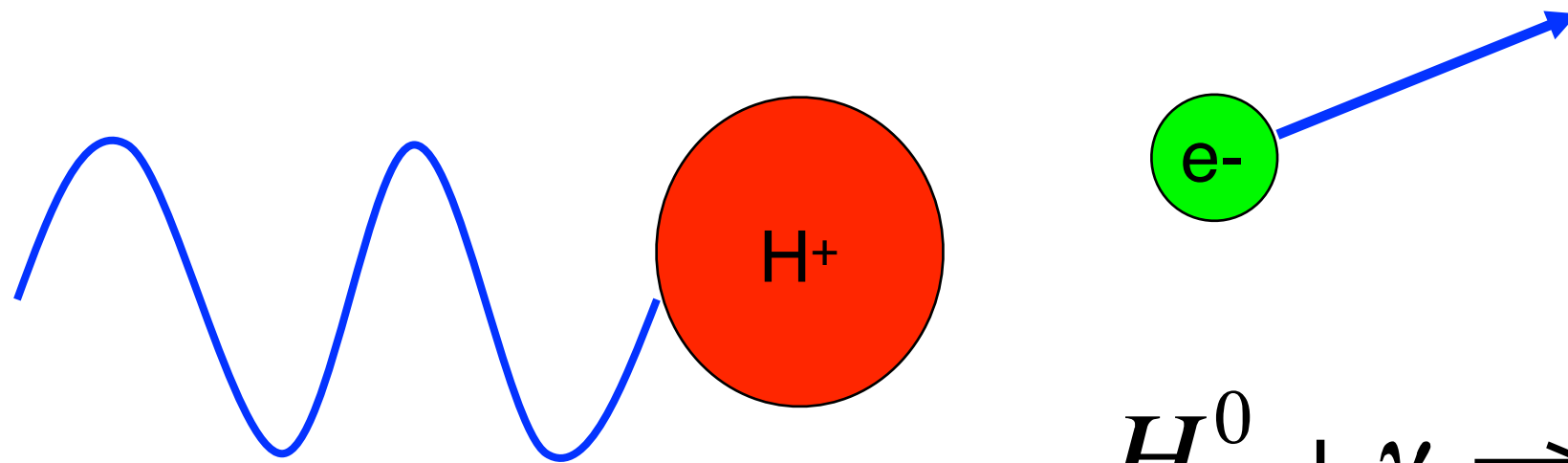


Irsic et al. (2023)

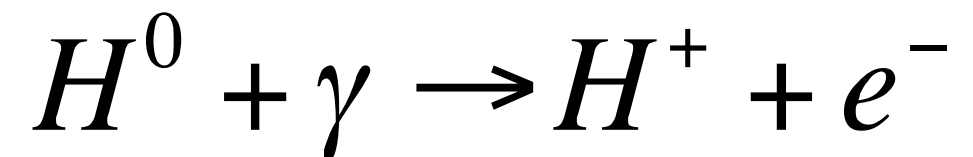
$$m_{\text{WDM}} > 5.7 \text{ keV} (2\sigma)$$

$$m_{\text{WDM}} > 5.1 \text{ keV} (2\sigma)$$

Heating by UV photons



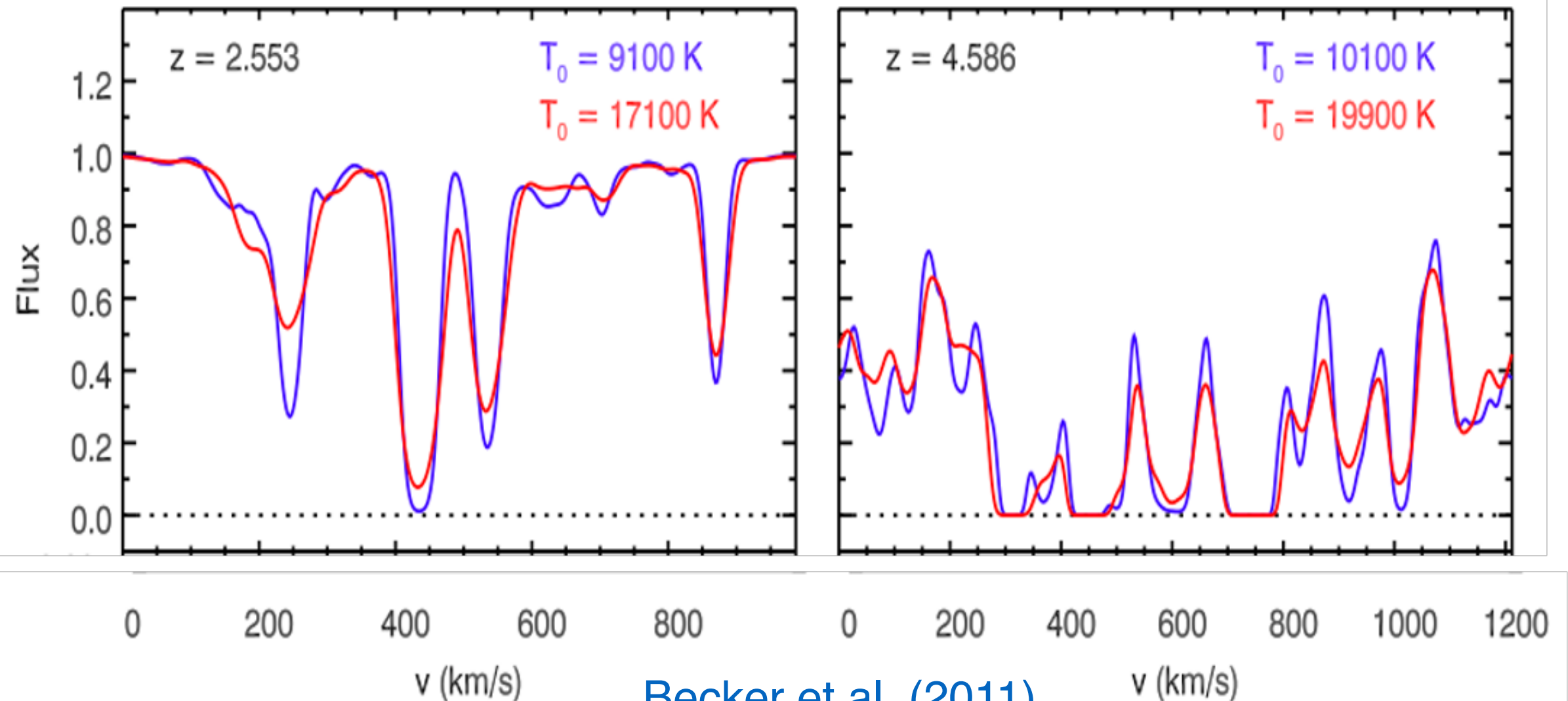
$E > 13.6 \text{ eV}$ photon



($E > 54.4 \text{ eV}$ photon for $\text{He}^+ + \gamma \rightarrow \text{He}^{2+} + e^-$)

Ejected photo-electrons interact with atoms / ions via scattering.

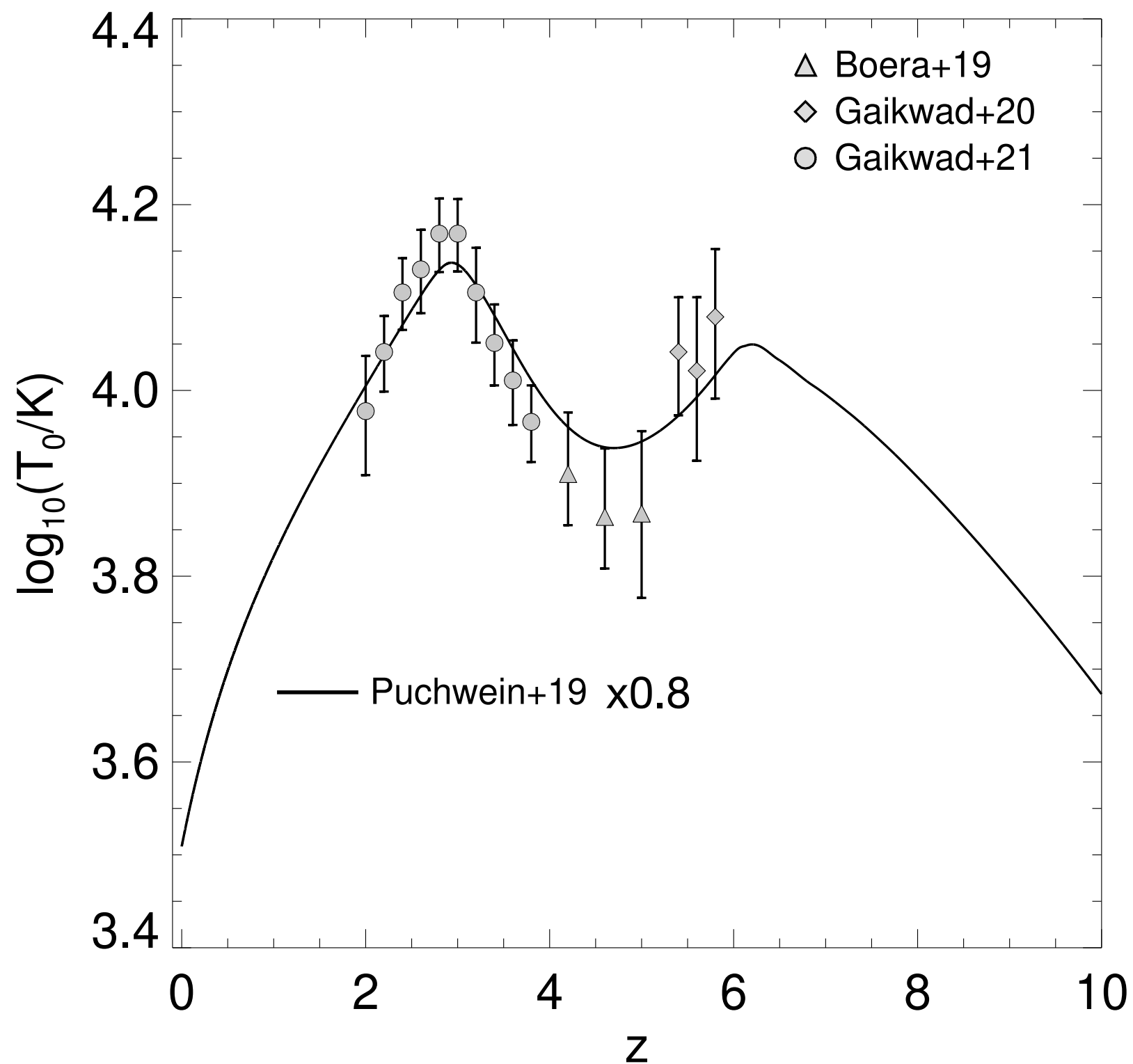
Lyman- α line widths



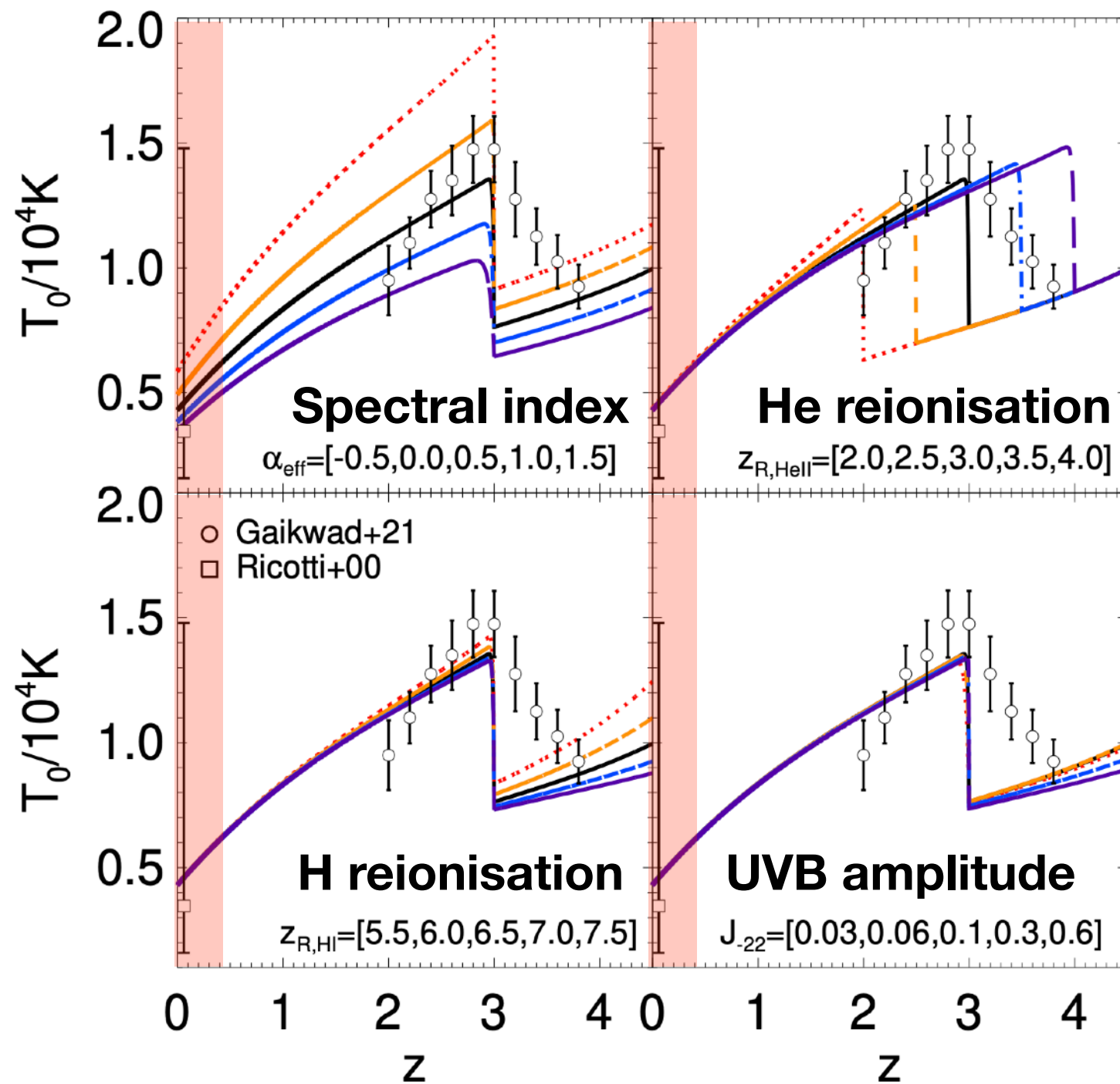
Becker et al. (2011)

$$b = \sqrt{\frac{2k_B T}{m_H}} = 12.9 \text{ km s}^{-1} \sqrt{\frac{T}{10^4 \text{ K}}}$$

Hydrogen temperature in IGM



Hydrogen temperature in the IGM



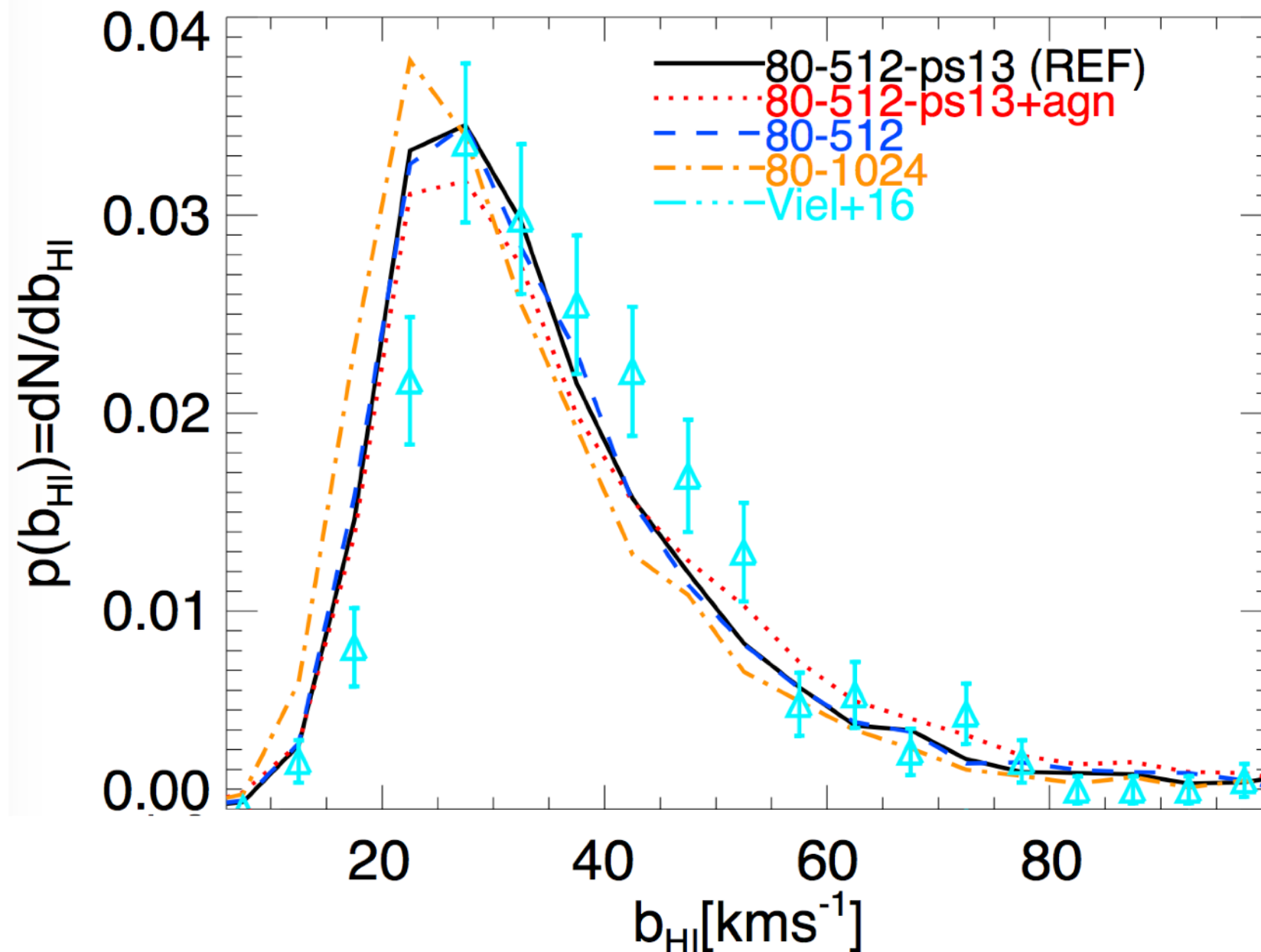
For power-law UV spectrum

$$J_\nu \propto \nu^{-\beta}$$

Temperature at low- z
 sensitive to spectral shape of
 UV background only

Bolton et al. (2022a)

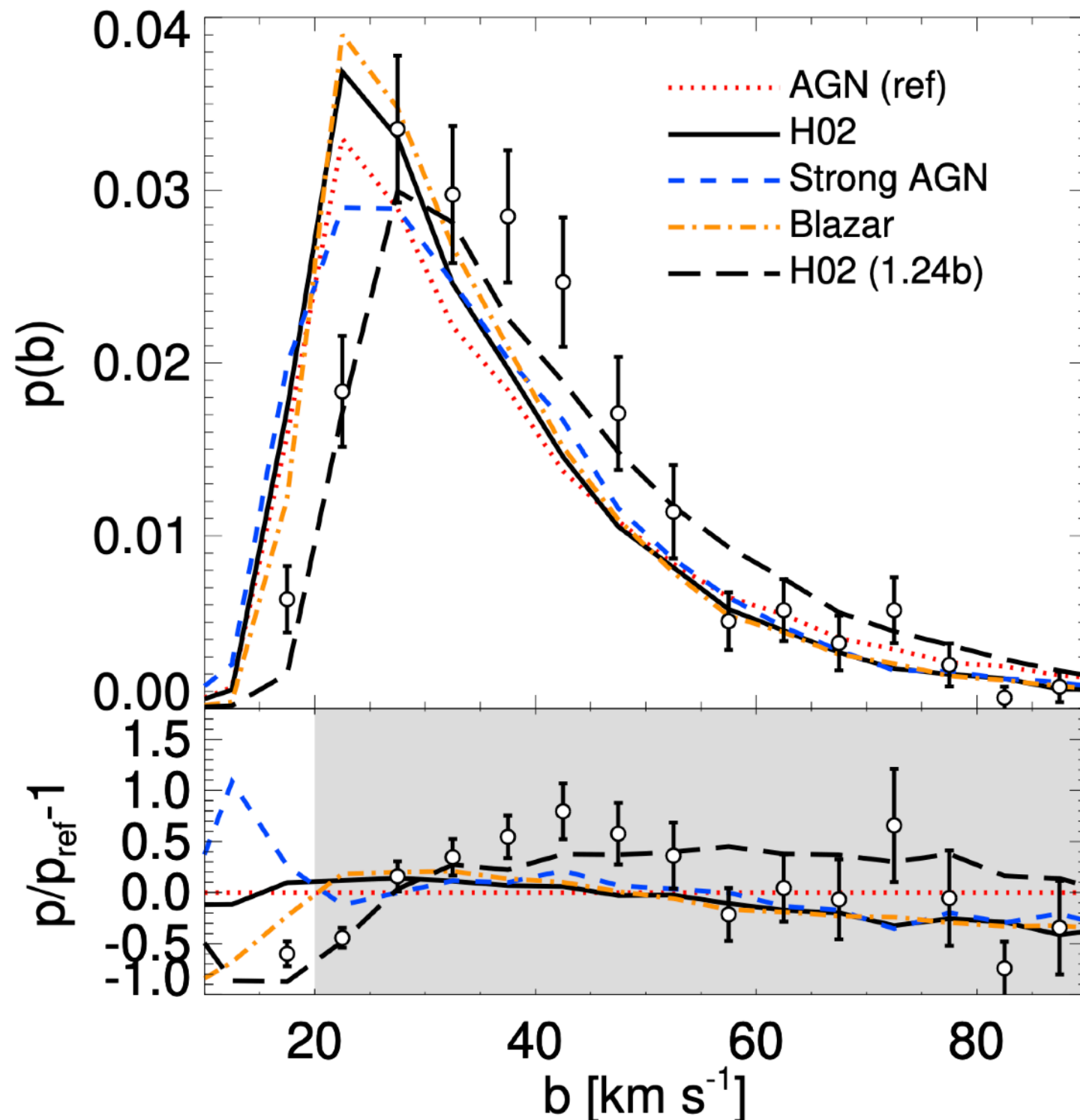
Line width distribution at $z \sim 0.1$



- Data at $0 < z < 0.2$ from 44 HST / COS spectra;
- Simulated line widths (with P-Gadget-3) are **too narrow** at $z=0.1$

Nasir et al. (2017)
see also Viel et al. (2017);

Line width distribution at $z \sim 0.1$



Bolton et al. (2022a)

Power-law UV spectrum

$$J_{\nu} \propto \nu^{-\beta}$$

Required by data (1σ)

$$\beta = -0.43^{+0.13}_{-0.26}$$

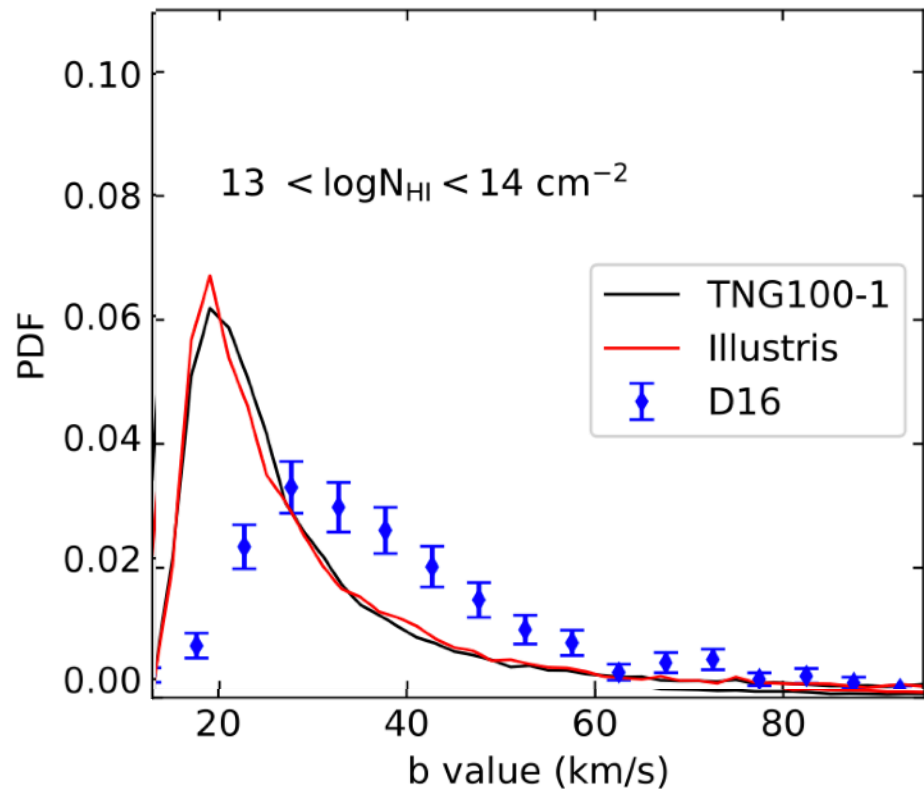
UVXB synthesis models

$$\beta \simeq 1.0$$

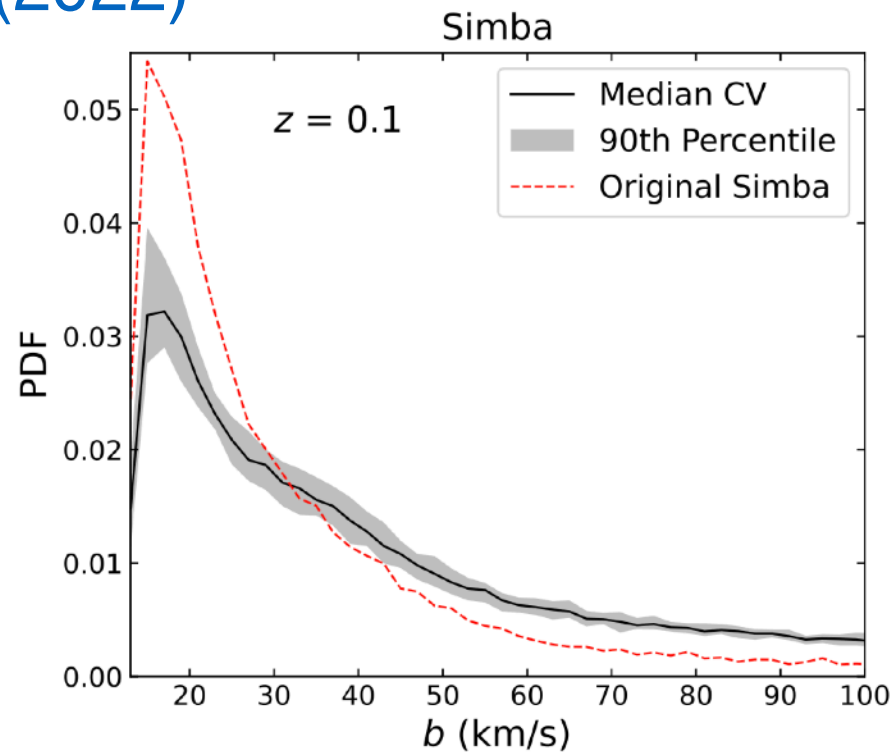
Typical QSO spectrum

$$\beta \simeq 1.5$$

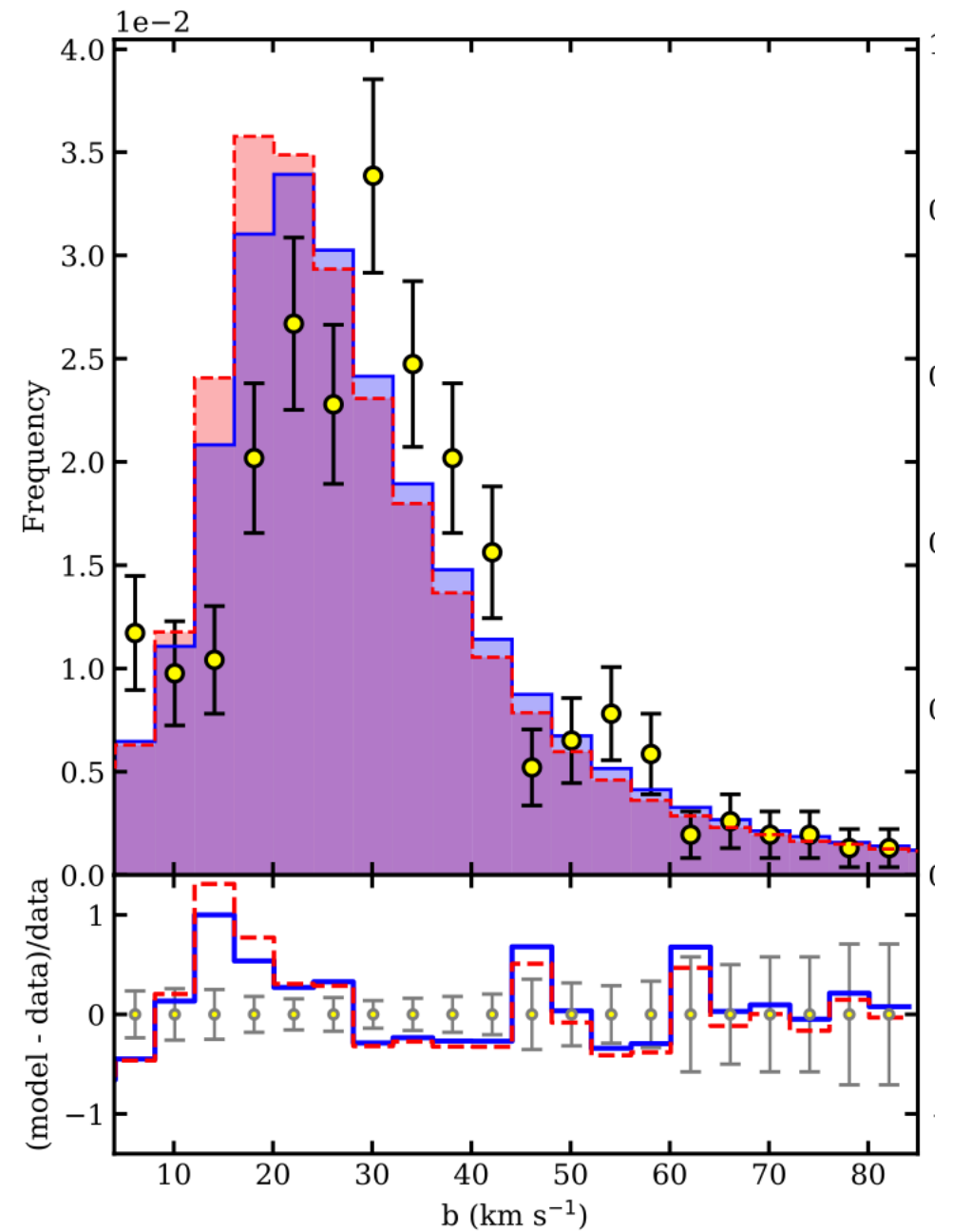
Other recent results



Burkhart et al. (2022)



Tillman et al. (2023)



Khaire et al. (2023)

Implications

1) Additional specific heat injection required from “non-canonical” source (i.e. not heating by stars or AGN) **at $z < 2.5$ only**

$$u \leq 6.9 \text{ eV } m_p^{-1}$$

Photo-electric emission by dust grains? (Inoue & Kamaya 2010).

But: requires $Z \geq 0.1 Z_\odot$ in low density IGM.

2) Unresolved line of sight turbulent velocity component driven by some persistent source of vorticity

$$v_{\text{turb}} \leq 8.5 \text{ km s}^{-1} (N_{\text{HI}} / 10^{13.5} \text{ cm}^{-2})^{0.21}$$

Speculative models for IGM turbulence exist (e.g. Evoli & Ferrara 2011)

But: increase simulation resolution, lines get narrower, not broader. Also, IGM is generally low density, quiescent gas.

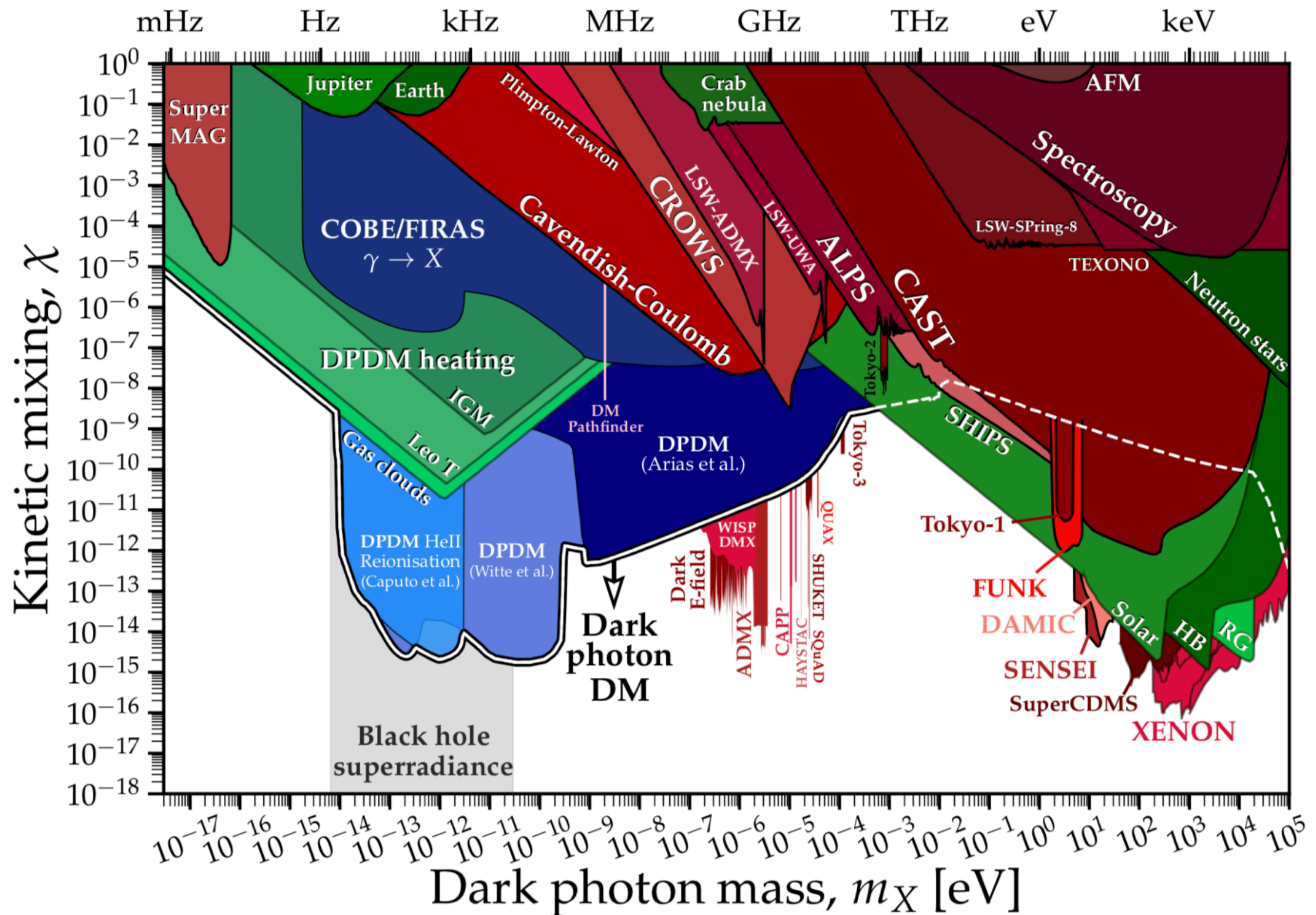
Bolton et al. (2022a)

Introduction to dark photons

- Dark photons: minimal and well-motivated extension of Standard Model which couples (i.e. converts) to the ordinary photon;
- Dark photons are a cold dark matter candidate;
- Key parameters are: **mass of the dark photon** and strength of the coupling (**kinetic mixing**) with the ordinary photon.

$$\mathcal{L}_{\gamma A'} = -\frac{1}{4}F_{\mu\nu}^2 - \frac{1}{4}(F'_{\mu\nu})^2 - \frac{\epsilon}{2}F^{\mu\nu}F'_{\mu\nu} + \frac{1}{2}m_{A'}^2(A'_\mu)^2$$

Overview of parameter space



Dark photon heating rate

- Conversion of dark photons to ordinary photons is resonantly enhanced when the local **effective plasma mass**, m_γ , of ordinary photons crosses the dark photon mass.

$$m_\gamma^2(z, \vec{x}) \simeq 1.4 \times 10^{-21} \text{ eV}^2 c^{-4} \left(\frac{n_e(z, \vec{x})}{\text{cm}^{-3}} \right)$$

- For IGM, the mean (comoving) electron density is $n_e \simeq 10^{-7} \text{ cm}^{-3}$
- Resonant conversion occurs in IGM for dark photon masses $m_{A'} \simeq 10^{-14} \text{ eV c}^{-2}$, yielding ordinary photons with frequency

$$\nu = m_{A'} c^2 / h \simeq 1 \text{ Hz}$$

Dark photon heating rate

- Low frequency photons undergo **free-free absorption** with a mean free path (e.g. Draine 2011)

$$\lambda_{\text{ff}} \simeq \frac{14 \text{ kpc}}{(1+z)^6} \Delta_{\text{b}}^{-2} \left(\frac{T}{10^4 \text{ K}} \right)^{3/2} \left(\frac{m_{A'}}{10^{-13} \text{ eV } c^{-2}} \right)^2$$

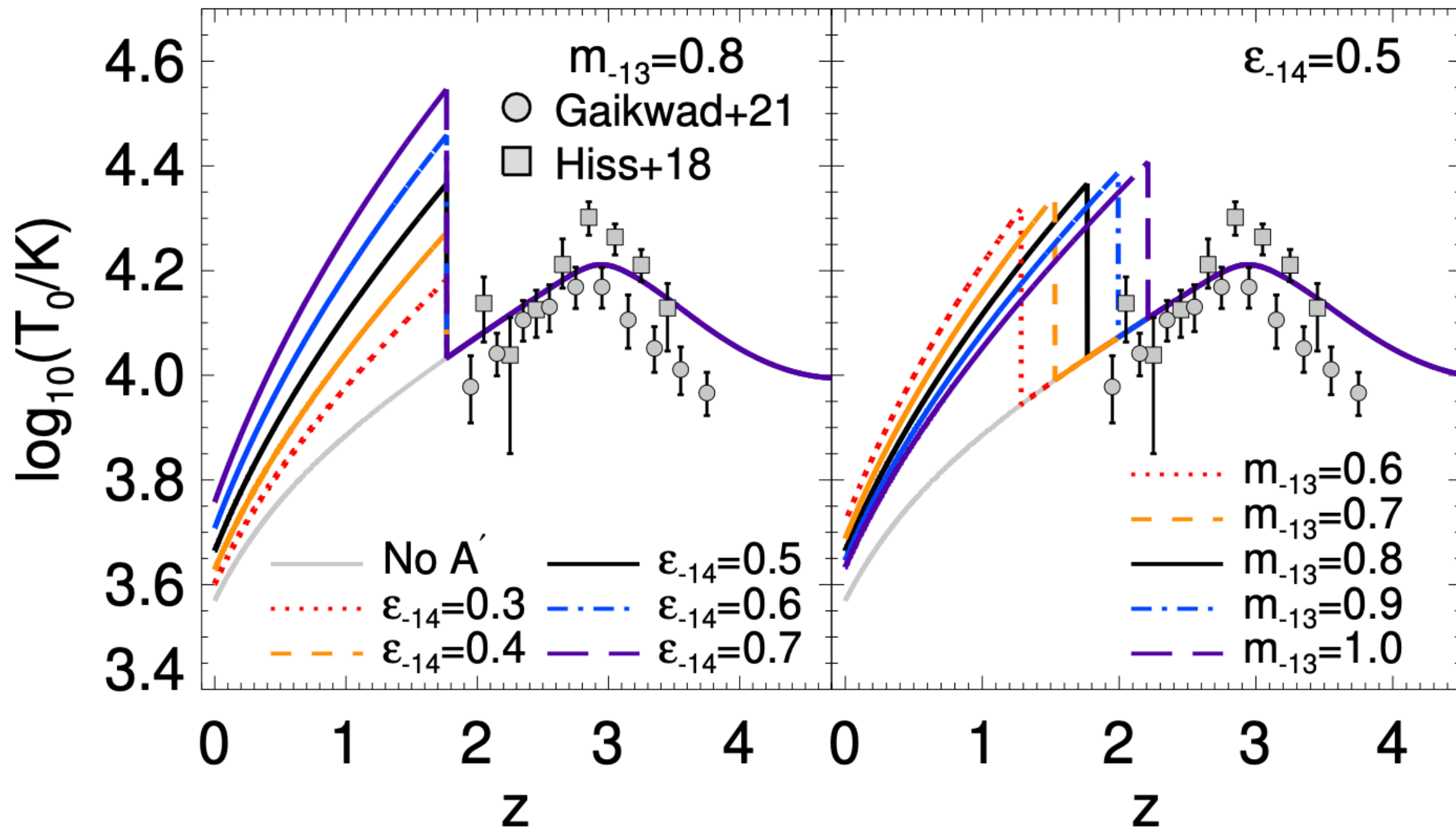
- Specific heat injection from free-free absorption

$$u \simeq 0.1 \text{ eV } m_{\text{p}}^{-1} \left(\frac{m_{A'}}{10^{-13} \text{ eV } c^{-2}} \right) \left(\frac{\epsilon}{10^{-15}} \right)^2 \left(\frac{3}{1+z_{\text{res}}} \right)^{3/2}$$

- Resonant conversion redshift depends on the local density

$$(1+z_{\text{res}}) \propto \Delta^{-1/3} \Rightarrow u \propto \Delta^{1/2}$$

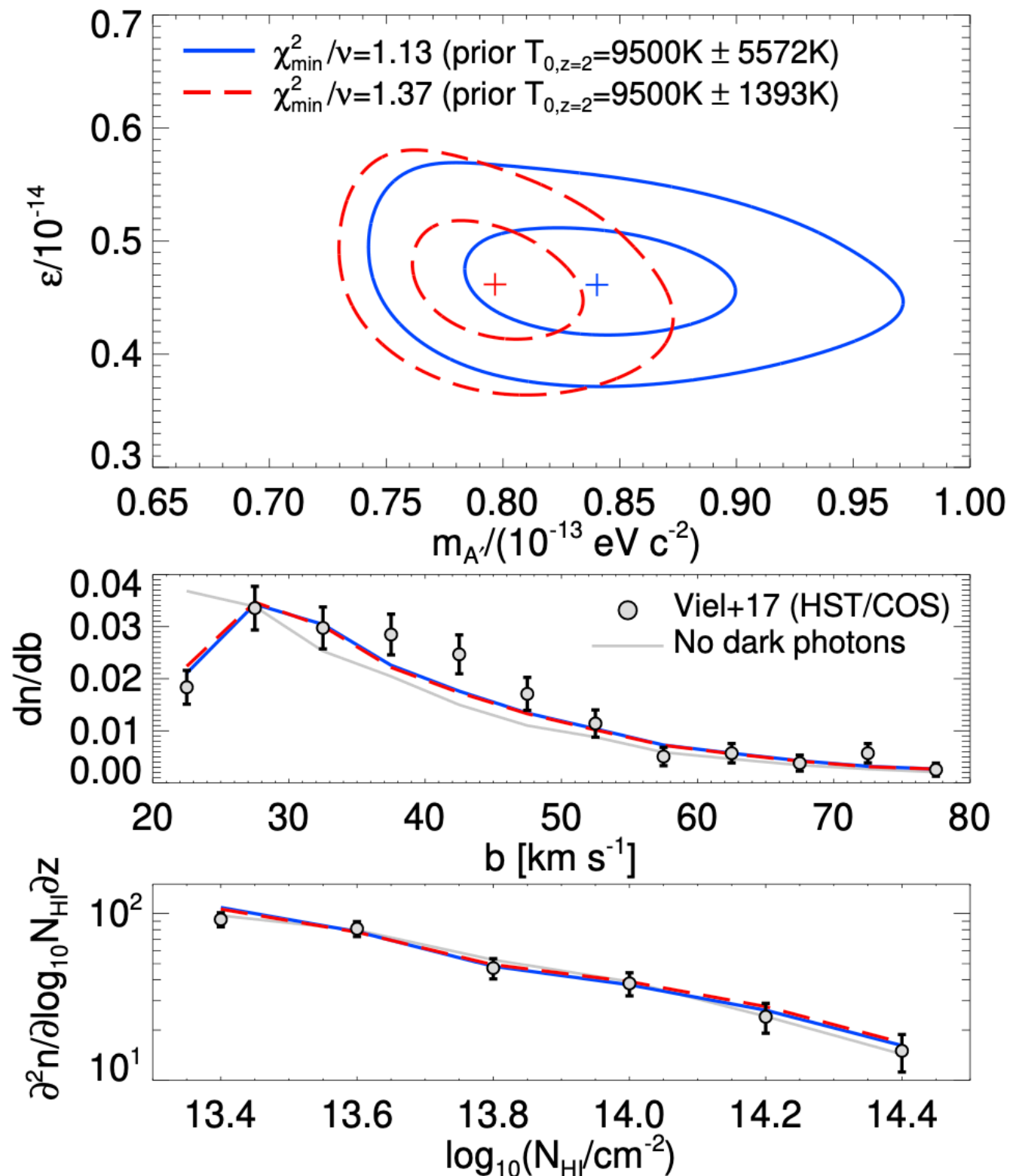
Dark photon heating rate



$$m_{-13} = \frac{m_{A'}}{10^{-13} \text{ eV c}^{-2}} \quad \epsilon_{-14} = \frac{\epsilon}{10^{-14}}$$

Bolton et al. (2022b)

Hints of dark photon dark matter?



- First implementation of DP heating in IGM hydrodynamical simulations;

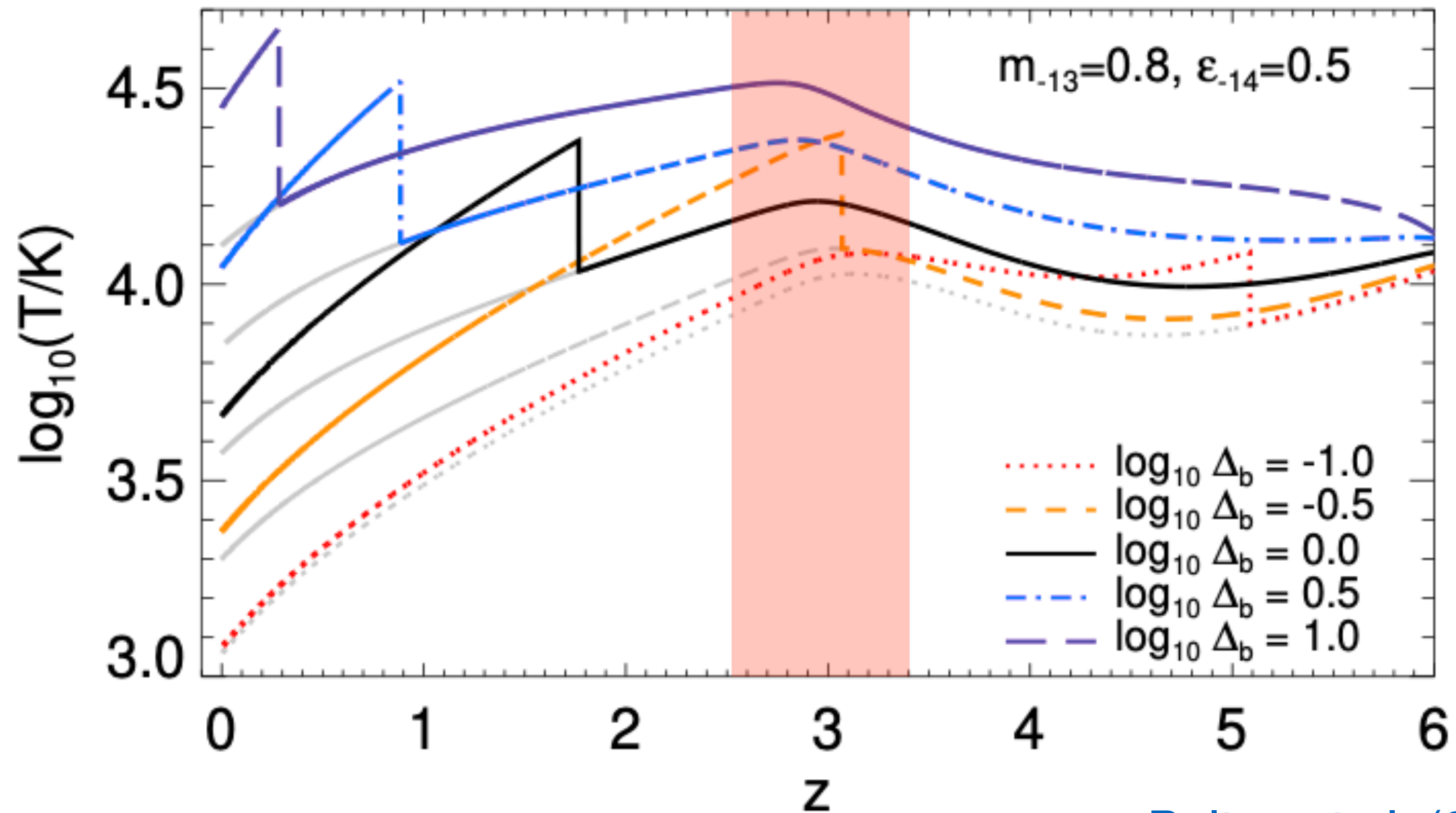
- Best fit for weak $T(z=2)$ prior

$$m_{A'} = 8.4 \pm 0.6 \times 10^{-14} \text{ eV c}^{-2}$$

$$\epsilon = 4.6^{+0.5}_{-0.4} \times 10^{-15}$$

Bolton et al. (2022b)

Dark photon heating with density



Bolton et al. (2022b)

- The resonant conversion redshift will also depend on the local density

$$(1 + z_{\text{res}}) \propto \Delta^{-1/3} \Rightarrow u \propto \Delta^{1/2}$$

Heating in voids

Mon. Not. R. Astron. Soc. **386**, 1131–1144 (2008)

doi:10.1111/j.1365-2966.2008.13114.x

Possible evidence for an inverted temperature–density relation in the intergalactic medium from the flux distribution of the Ly α forest

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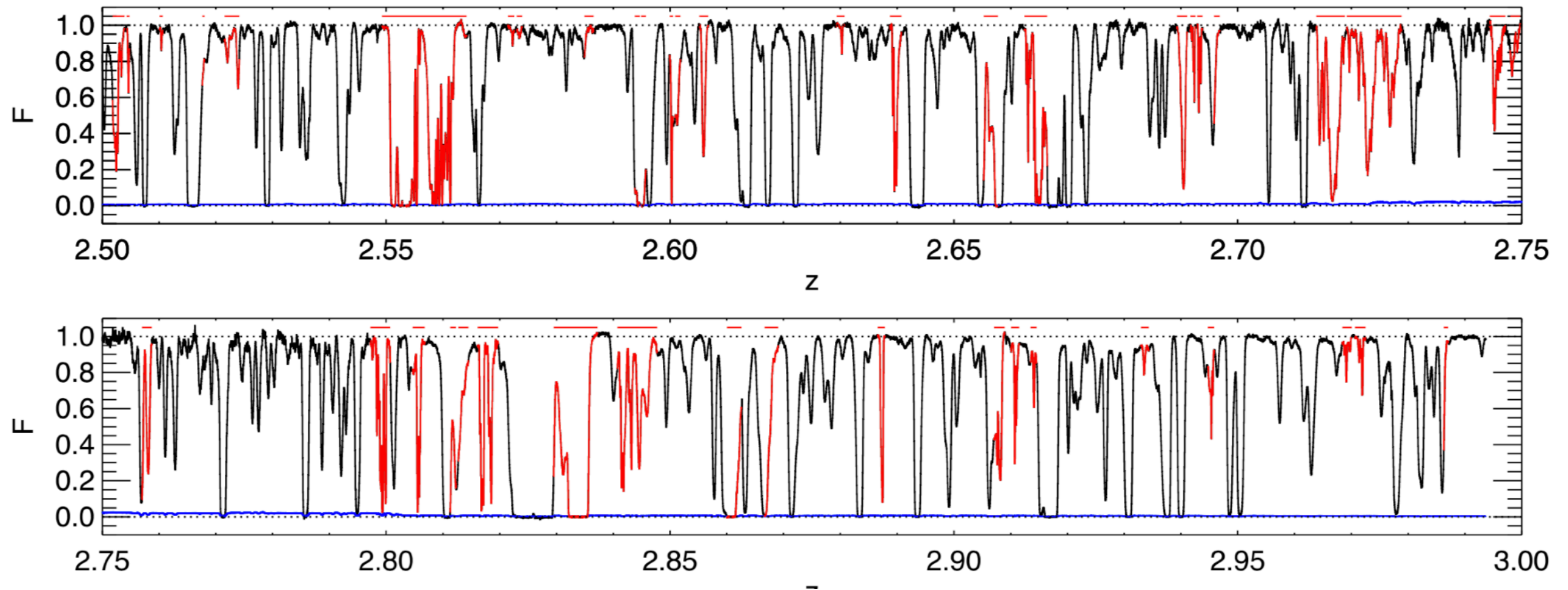
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ABSTRACT

We compare the improved measurement of the Ly α forest flux probability distribution at $1.7 < z < 3.2$ presented by Kim et al. to a large set of hydrodynamical simulations of the Ly α forest with different cosmological parameters and thermal histories. The simulations are in good agreement with the observational data if the temperature–density relation for the low-density intergalactic medium (IGM), $T = T_0 \Delta^{\gamma-1}$, is either close to isothermal or inverted ($\gamma < 1$). Our results suggest that the voids in the IGM may be significantly hotter and the thermal state of the low-density IGM may be substantially more complex than is usually assumed at these redshifts. We discuss radiative transfer effects which alter the spectral shape of ionizing radiation during the epoch of He II reionization as a possible physical mechanism for achieving an inverted temperature–density relation at $z \simeq 3$.

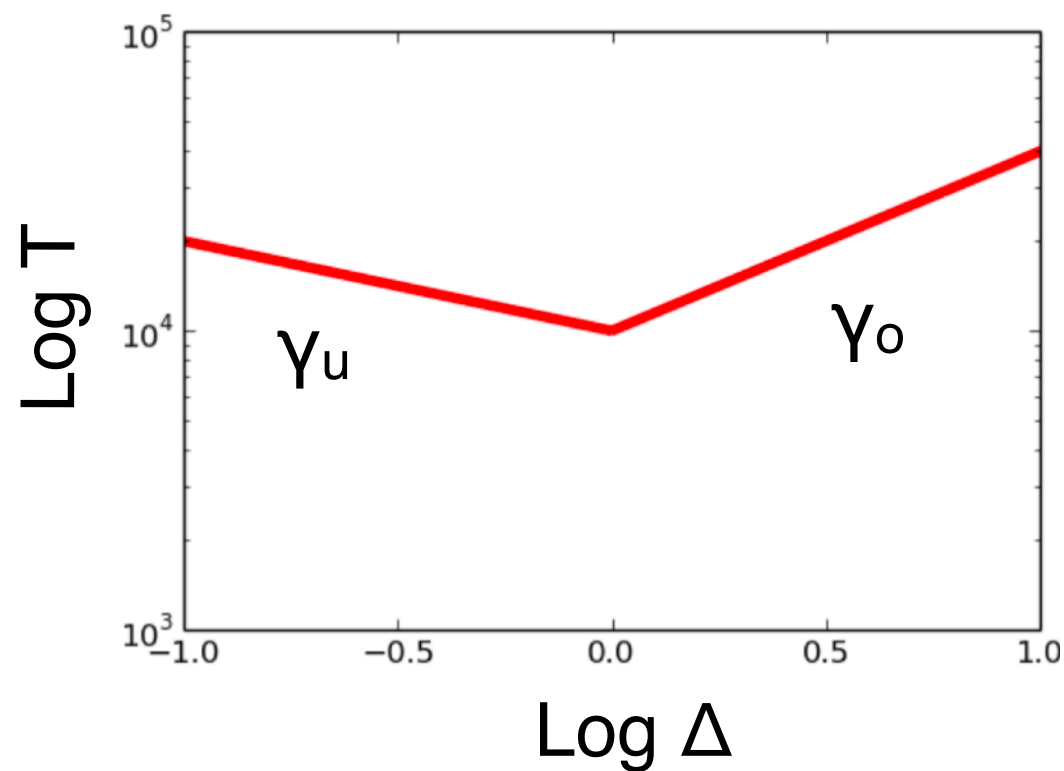
DEEP spectrum



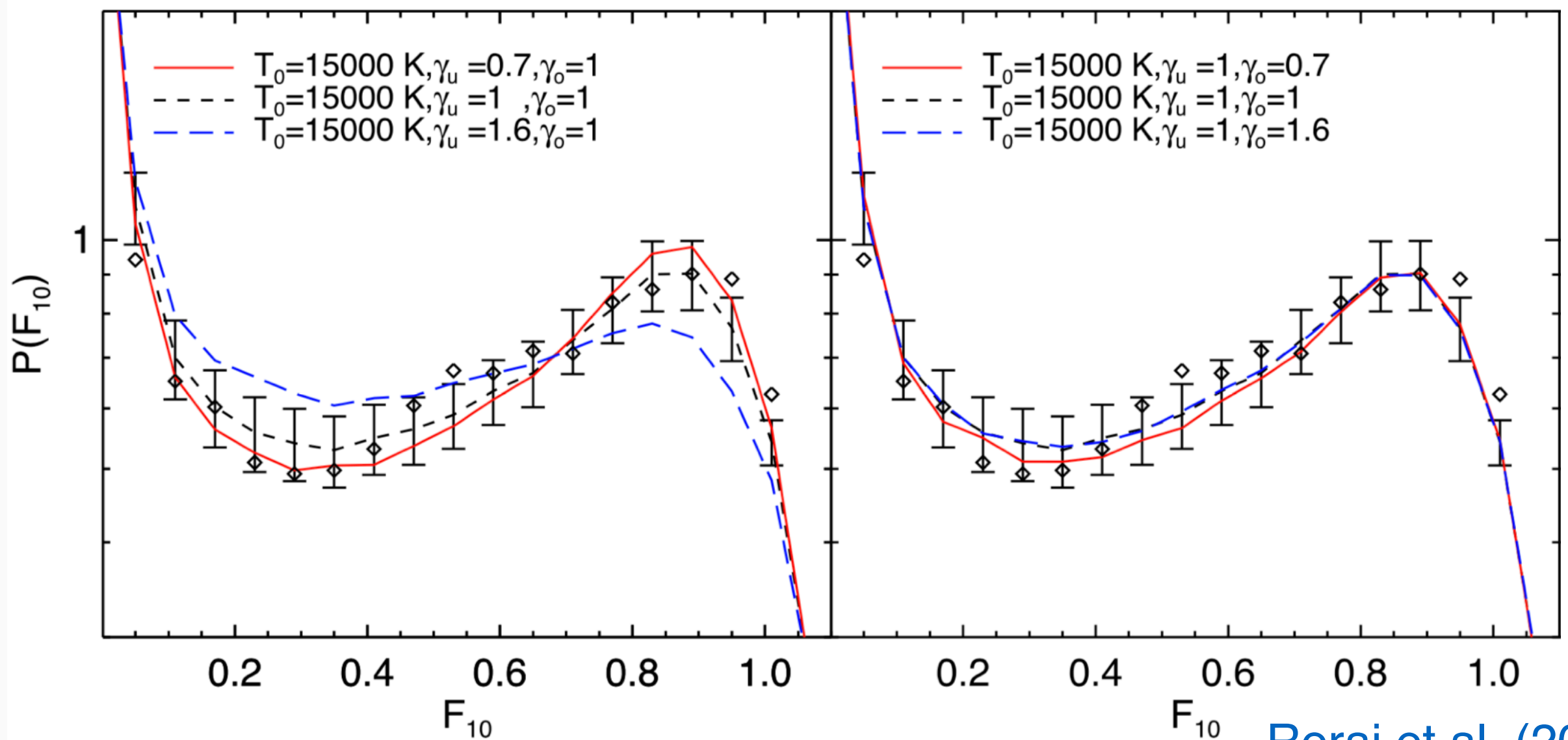
Rorai et al. (2017)

HE0940-1050 ($z=3.09$)

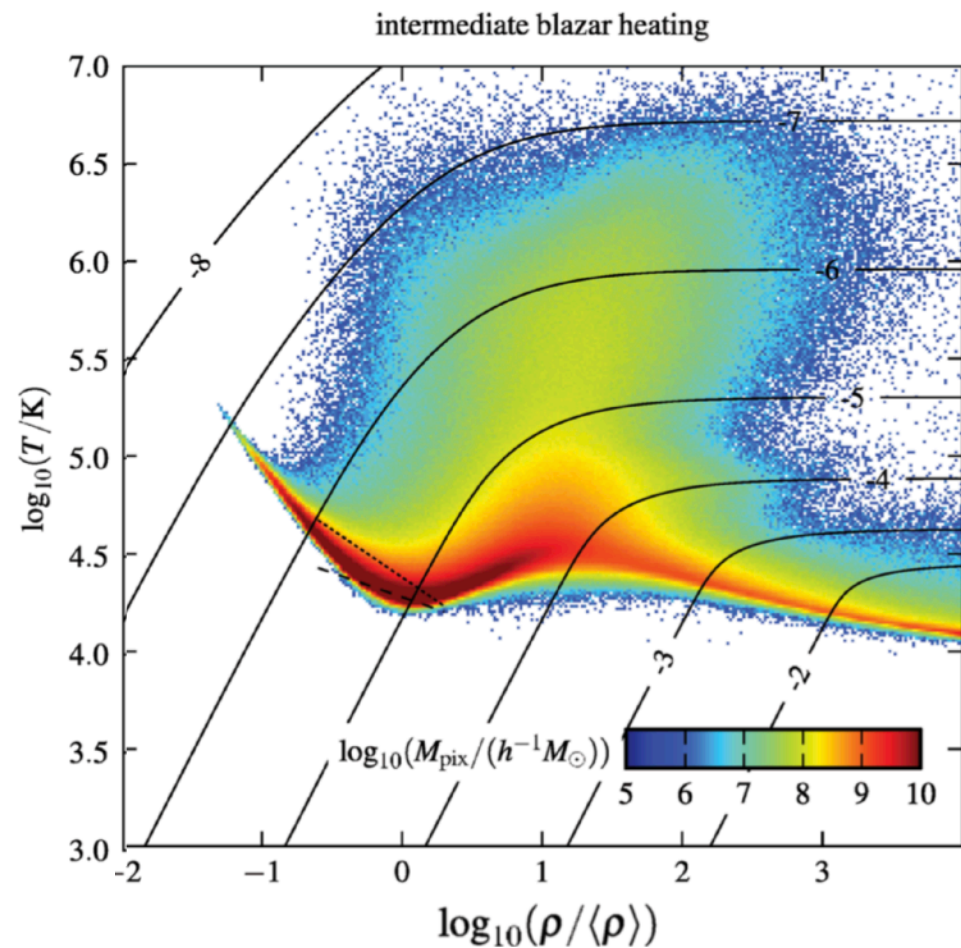
- 64.4 hrs on UVES (PI S. Cristiani)
- $R \sim 45000$
- $S/N=280$ per resolution element



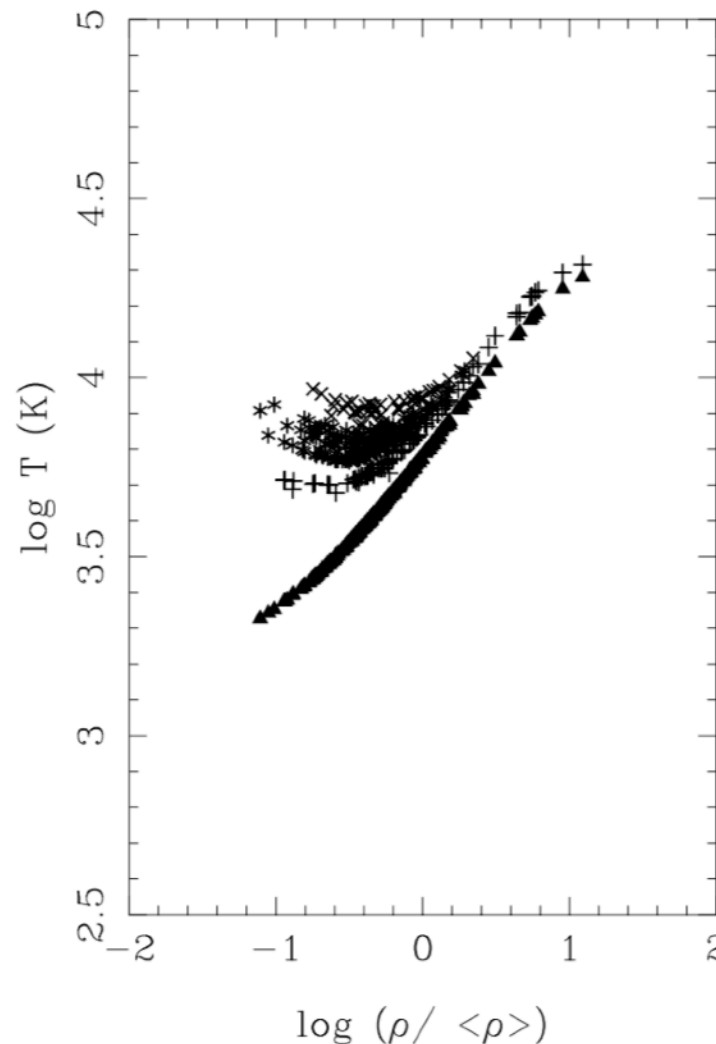
- Temperature-density relation isothermal/inverted for underdense gas (90% confidence)



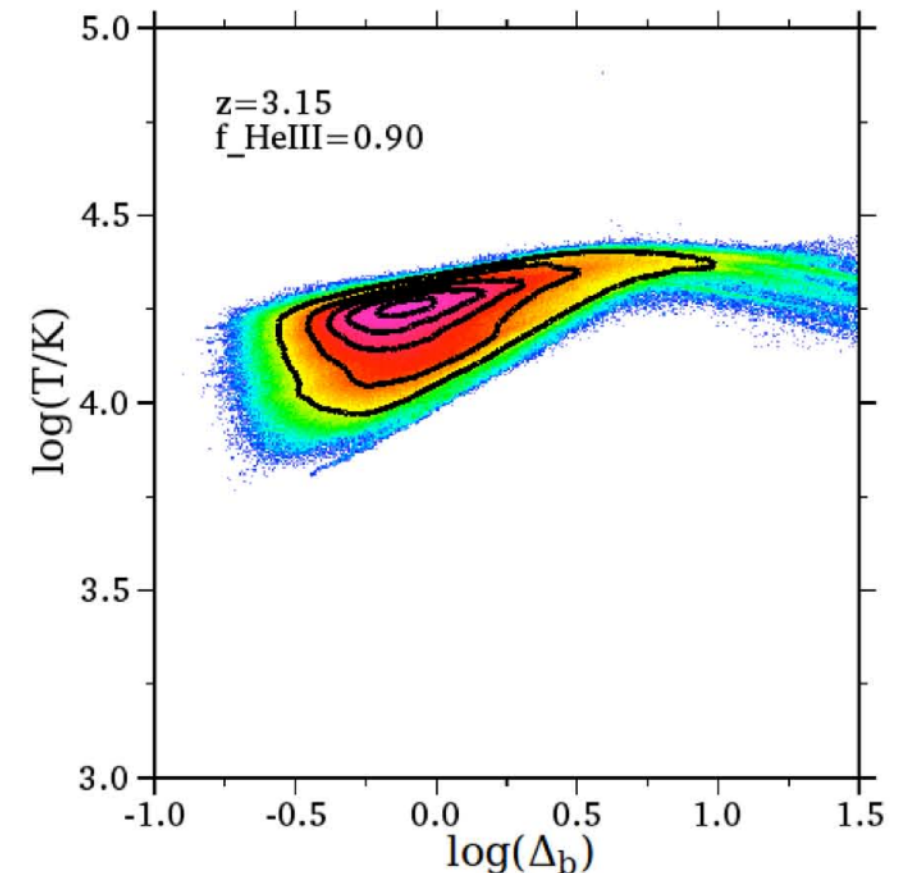
Generating hot voids at $z \sim 3$



Puchwein+12



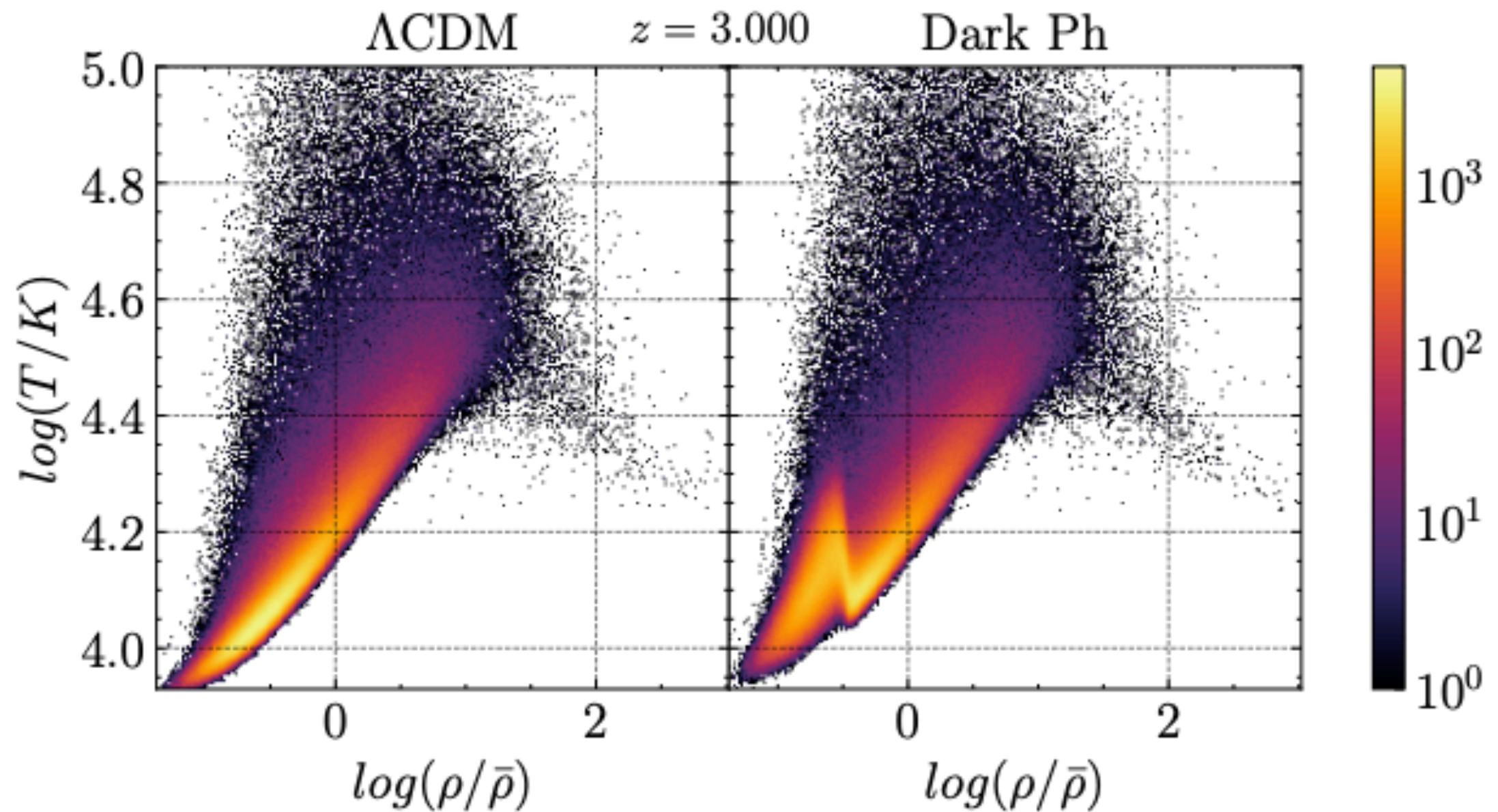
Bolton+04



Compostella+13

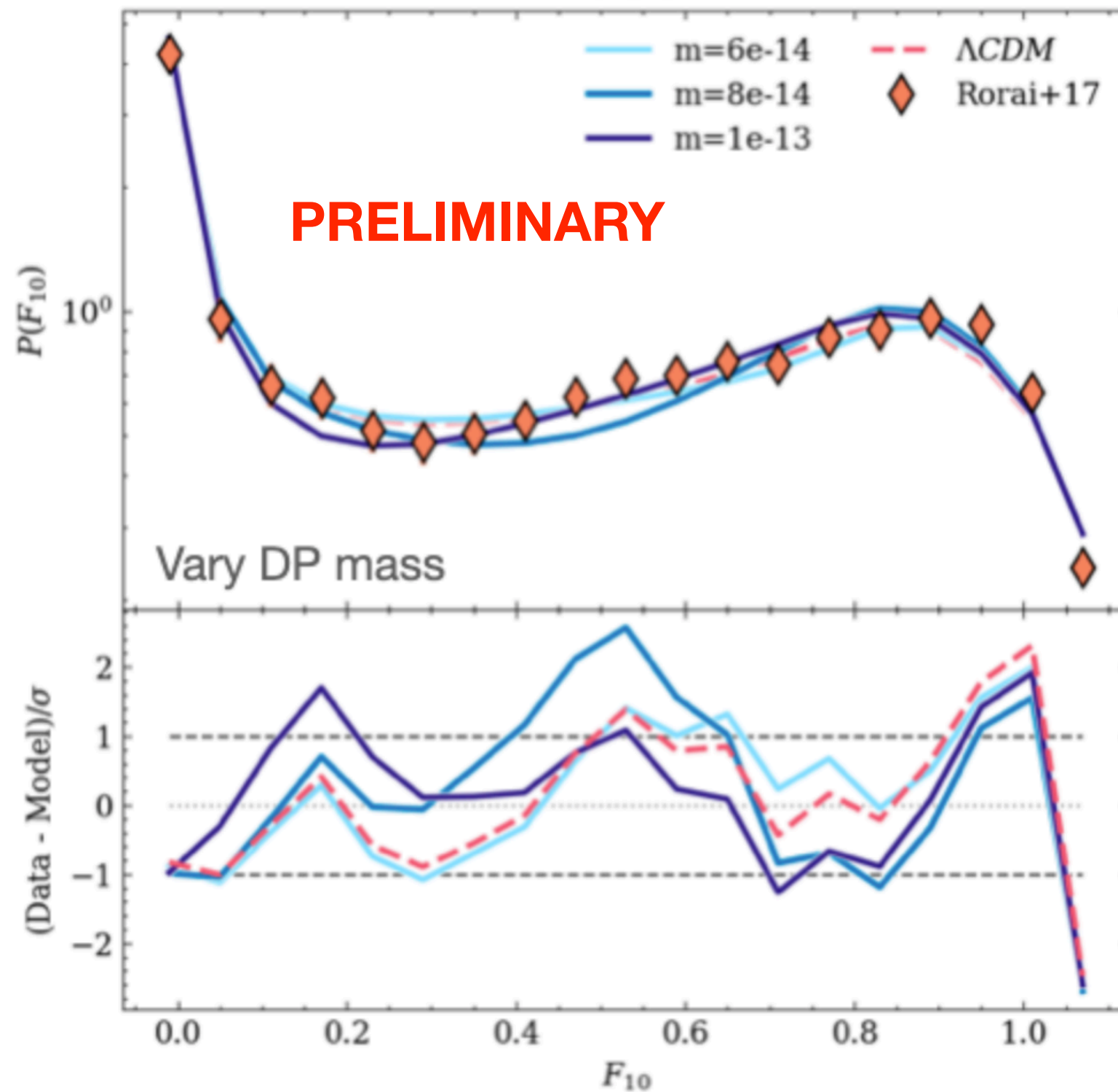
- Volumetric heating from blazar TeV emission heats under-dense gas (but see [Miniati & Elyiv 2013](#))
- Patchy heating and spectral filtering induces scatter (H-I *and* He-II reionisation)

Best fit dark photon model at $z=3$



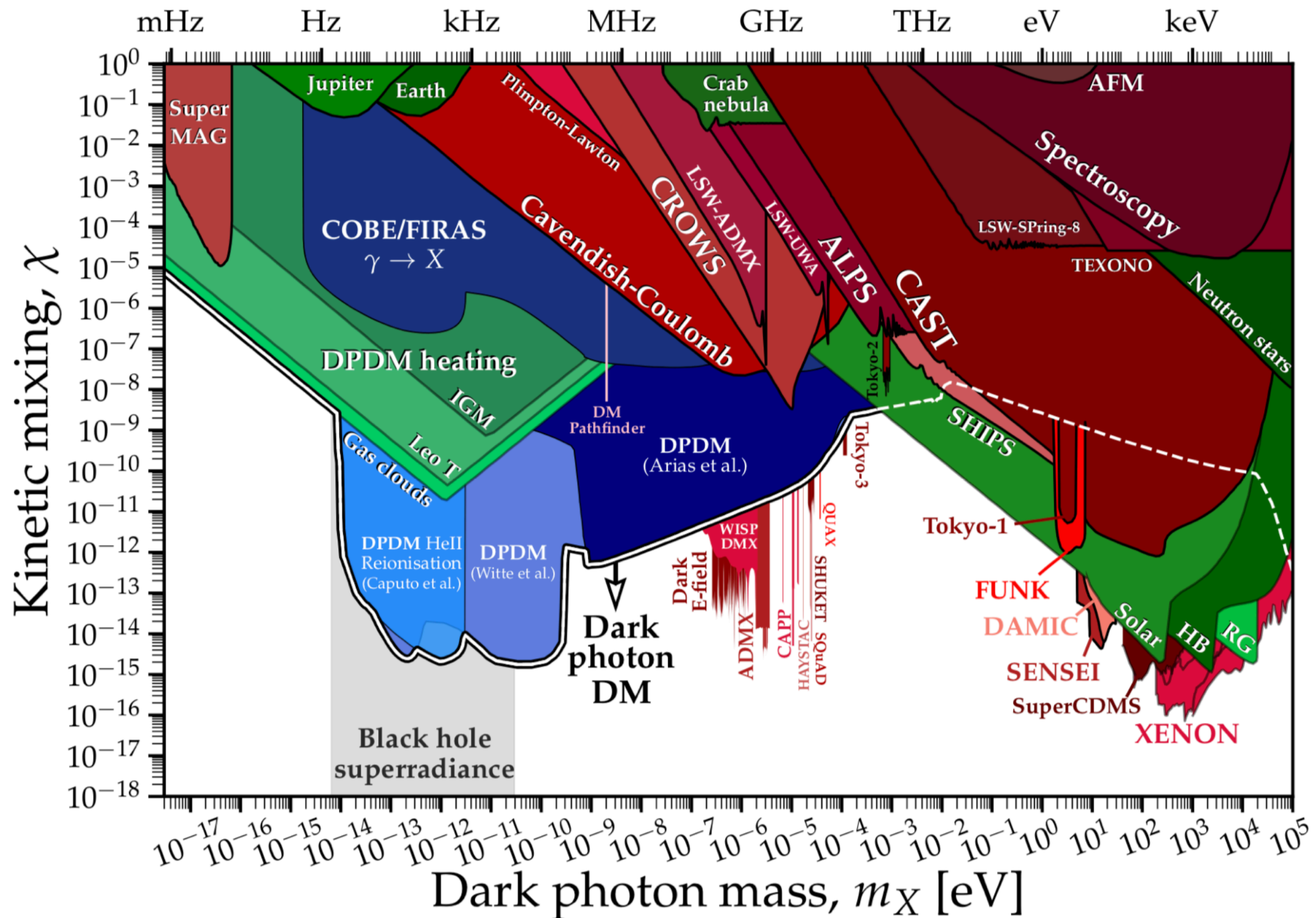
Trost et al. (in prep.)

Best fit dark photon model at $z=3$



Trost et al. (in prep.)

Overview of parameter space



Summary

- **Lyman- α forest ($N_{\text{HI}} \sim 10^{13.3} - 10^{14.5} \text{ cm}^{-2}$) at $z \sim 0.1$**
 - Hydro-simulations struggle to reproduce line widths at $z=0.1$;
 - Reproducing line width distribution with additional photo-heating requires unphysically hard UV spectrum;
 - AGN/stellar feedback has minimal impact on line widths;
 - Unresolved turbulence? Non-canonical heating? Currently unsolved.
- **Dark photons**
 - First hydro-simulations that self-consistently follow heating from resonant conversion of dark photons;
 - Observed $z=0.1$ line width discrepancy alleviated by dark photons, while preserving good agreement with models and data at $z > 2$
 - Next steps: test for preferred model at $z=0.1$ on voids at $z=3$. Can we improve on existing dark photon parameter constraints?